

Healthcare Operations & Financial Overview

1. Project Overview

Hospital operations and financial billing are frequently managed in separate systems, making it difficult for administrators to evaluate how daily patient flow impacts overall hospital revenue. This project resolves that gap by bringing operational throughput and billing performance together into a single Power BI dashboard.

The resulting dashboard allows hospital directors, CFOs, and operational leads to monitor patient admissions, bed occupancy durations, clinical drivers, and insurance payer distributions from one place

2. Dataset Summary

- Rows: 54,966
- Columns: 15
- Key Features:
 - Patient Demographics: Patient Name, Age, Gender, Blood Type
 - Clinical & Admission Details: Medical Condition, Admission Type, Date of Admission, Length of Stay
 - Hospital & Operational Data: Hospital Branch, Attending Doctor
 - Financial & Payer Information: Insurance Provider, Billing Amount

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the 54,966 patient admission records into a Data Frame using `pandas`

```
df = pd.read_csv("healthcare_dataset.csv")

df
```

	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
0	Bobby JacksOn	30	Male	B-	Cancer	2024-01-31	Matthew Smith	Sons and Miller	Blue Cross	18856.281306	328	Urgent	2024-02-02	Paracetamol	Normal
1	LesLie TErRy	62	Male	A+	Obesity	2019-08-20	Samantha Davies	Kim Inc	Medicare	33643.327287	265	Emergency	2019-08-26	Ibuprofen	Inconclusive
2	DaNnY sMitH	76	Female	A-	Obesity	2022-09-22	Tiffany Mitchell	Cook PLC	Aetna	27955.096079	205	Emergency	2022-10-07	Aspirin	Normal
3	andrEw waTtS	28	Female	O+	Diabetes	2020-11-18	Kevin Wells	Hernandez Rogers and Vang,	Medicare	37909.782410	450	Elective	2020-12-18	Ibuprofen	Abnormal
4	adrIENNE bEll	43	Female	AB+	Cancer	2022-09-19	Kathleen Hanna	White-White	Aetna	14238.317814	458	Urgent	2022-10-09	Penicillin	Abnormal
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
55495	eLIZABeTH jaCkSOOn	42	Female	O+	Asthma	2020-08-16	Joshua Jarvis	Jones-Thompson	Blue Cross	2650.714952	417	Elective	2020-09-15	Penicillin	Abnormal
55496	KYle pEReZ	61	Female	AB-	Obesity	2020-01-23	Taylor Sullivan	Tucker-Moyer	Cigna	31457.797307	316	Elective	2020-02-01	Aspirin	Normal
55497	HEATHer WaNG	38	Female	B+	Hypertension	2020-07-13	Joe Jacobs DVM	and Mahoney Johnson Vasquez,	UnitedHealthcare	27620.764717	347	Urgent	2020-08-10	Ibuprofen	Abnormal
55498	JENnIFER JOnES	43	Male	O-	Arthritis	2019-05-25	Kimberly Curry	Jackson Todd and Castro,	Medicare	32451.092358	321	Elective	2019-05-31	Ibuprofen	Abnormal
55499	jAMES GARCIA	53	Female	O+	Arthritis	2024-04-02	Dennis Warren	Henry Sons and	Aetna	4010.134172	448	Urgent	2024-04-29	Ibuprofen	Abnormal

55500 rows x 15 columns

- Initial Exploration: Used `df.head()` to get a basic overview about the Columns and its Rows, `.describe()` for summary statistics

	Age	Billing Amount	Room Number
count	55500.000000	55500.000000	55500.000000
mean	51.539459	25539.316097	301.134829
std	19.602454	14211.454431	115.243069
min	13.000000	-2008.492140	101.000000
25%	35.000000	13241.224652	202.000000
50%	52.000000	25538.069376	302.000000
75%	68.000000	37820.508436	401.000000
max	89.000000	52764.276736	500.000000

	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
0	Bobby JacksOn	30	Male	B-	Cancer	2024-01-31	Matthew Smith	Sons and Miller	Blue Cross	18856.281306	328	Urgent	2024-02-02	Paracetamol	Normal
1	LesLie TErRy	62	Male	A+	Obesity	2019-08-20	Samantha Davies	Kim Inc	Medicare	33643.327287	265	Emergency	2019-08-26	Ibuprofen	Inconclusive
2	DaNnY sMitH	76	Female	A-	Obesity	2022-09-22	Tiffany Mitchell	Cook PLC	Aetna	27955.096079	205	Emergency	2022-10-07	Aspirin	Normal
3	andrEw waTtS	28	Female	O+	Diabetes	2020-11-18	Kevin Wells	Hernandez Rogers and Vang,	Medicare	37909.782410	450	Elective	2020-12-18	Ibuprofen	Abnormal
4	adrIENNE bEll	43	Female	AB+	Cancer	2022-09-19	Kathleen Hanna	White-White	Aetna	14238.317814	458	Urgent	2022-10-09	Penicillin	Abnormal

- **Checking for Null & Duplicate Value:** Used `df.duplicated().sum()` to check the total number of duplicate row in the Data set, and `df.isna().sum()` to check null values

```
print(df.duplicated().sum()) # 534 Duplicate Rows

print(df.isna().sum()) # 0 null values in the data set
```



```
534
Name          0
Age           0
Gender        0
Blood Type    0
Medical Condition 0
Date of Admission 0
Doctor        0
Hospital      0
Insurance Provider 0
Billing Amount 0
Room Number   0
Admission Type 0
Discharge Date 0
Medication    0
Test Results  0
dtype: int64
```

- **Data Cleaning:** Handled missing values, Purged duplicates (`drop_duplicates()`), standardized date fields, and validated billing figures.

```
df = df.drop_duplicates()

df = df.reset_index(drop = True) # Reset the Index

print(df.duplicated().sum()) # there are 0 duplicate Rows in the Dataset

0

print(f"Cleaned Row Count: {len(df)}")

Cleaned Row Count: 54966
```

• Standardization:

5. Standardization

```
print(df.sample(10).to_string())
```

	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
26423	suSAN FreEMan	56	Male	AB+	Obesity	2019-10-20	Daniel Jones	Shaw Ltd	Medicare	26689.909282	200	Elective	2019-11-05	Lipitor	Inconclusive
34409	rOBert ayAla	44	Male	AB-	Hypertension	2021-09-30	Nancy Smith	Alexander Patton, Lowery and	Aetna	39252.176546	142	Urgent	2021-10-13	Aspirin	Inconclusive
13719	sARah fishEr	69	Female	A+	Hypertension	2021-01-18	Eric Massey	and Wilson, Wright Moore	Aetna	19369.825313	258	Elective	2021-02-02	Ibuprofen	Normal
18520	aLEXa WarD	35	Female	AB+	Hypertension	2023-01-22	Jessica Russell	Group Chavez	Medicare	2507.353574	425	Emergency	2023-01-28	Aspirin	Normal
21524	juLIe beARd	77	Female	AB+	Cancer	2020-06-11	Ashley Schmidt	and Peterson, Taylor Bridges	Cigna	8946.712555	222	Urgent	2020-06-18	Penicillin	Normal
33269	jaMES WALtOn	33	Female	O+	Diabetes	2022-11-30	Jamie Powell	Foster and Jefferson, Herring	Aetna	9773.210535	219	Elective	2022-12-12	Paracetamol	Normal
3558	tINa MoNTes	26	Female	AB-	Cancer	2022-05-17	Allison Jackson	Ward and Christensen, Russell	UnitedHealthcare	8308.032812	329	Emergency	2022-06-10	Penicillin	Inconclusive
3595	luCAs SchwaRtZ	43	Female	A-	Obesity	2022-10-01	Leah Simmons	and Davis Blankenship McDowell,	Medicare	38082.474571	264	Emergency	2022-10-12	Lipitor	Inconclusive
12147	jaSoN coLe	84	Male	A-	Arthritis	2023-04-05	Julia Rogers	Rosales-Gonzalez	Blue Cross	35082.700435	429	Elective	2023-04-08	Penicillin	Inconclusive
14711	JACOB rICHaRdSon	19	Male	O-	Arthritis	2024-04-13	Sean Johnson	and Young, Carpenter Henry	Aetna	45396.088549	295	Urgent	2024-05-12	Paracetamol	Normal

Generate Code Markdown

```
df["Name"] = df["Name"].str.strip().str.title() # Now the Values ie Rows in Column "Name" Looks standardized
```

```
df["Doctor"] = df["Doctor"].str.strip().str.title()
```

```
df["Hospital"] = df["Hospital"].str.strip()
```

Decimal Value Handling

```
df["Billing Amount"] = df["Billing Amount"].round(2) # Keeps only 2 Digits after the Decimal
```

```
print(df.head().to_string())
```

	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
0	Bobby Jackson	30	Male	B-	Cancer	2024-01-31	Matthew Smith	Sons and Miller	Blue Cross	18856.28	328	Urgent	2024-02-02	Paracetamol	Normal
1	Leslie Terry	62	Male	A+	Obesity	2019-08-20	Samantha Davies	Kim Inc	Medicare	33643.33	265	Emergency	2019-08-26	Ibuprofen	Inconclusive
2	Danny Smith	76	Female	A-	Obesity	2022-09-22	Tiffany Mitchell	Cook PLC	Aetna	27955.10	205	Emergency	2022-10-07	Aspirin	Normal
3	Andrew Watts	28	Female	O+	Diabetes	2020-11-18	Kevin Wells	Hernandez Rogers and Vang,	Medicare	37909.78	450	Elective	2020-12-18	Ibuprofen	Abnormal
4	Adrienne Bell	43	Female	AB+	Cancer	2022-09-19	Kathleen Hanna	White-White	Aetna	14238.32	458	Urgent	2022-10-09	Penicillin	Abnormal

- **Standardizing the Column Names:** Used `df.columns.str.replace(" ", "-")` to replace the white spaces between the word with “-”

```
df.columns = df.columns.str.replace(" ", "-")

print(df.columns)
```

[32]

```
Index(['Name', 'Age', 'Gender', 'Blood_Type', 'Medical_Condition',
       'Date_of_Admission', 'Doctor', 'Hospital', 'Insurance_Provider',
       'Billing_Amount', 'Room_Number', 'Admission_Type', 'Discharge_Date',
       'Medication', 'Test_Results'],
      dtype='str')
```

- **Feature Engineering:**

8. Feature Engineering

1. Creating a New Columns that answers the Question "How many Days did the patient Stay"

```
df["Lenght_of_Stay"] = (df["Discharge_Date"] - df["Date_of_Admission"]).dt.days
```

[]

```
df = df.rename(columns={
    "Lenght_of_Stay" : "Days_Stayed"
})
```

55]

2. Creating a new Flag Column based on the Column "Billing_Amount"

```
conditions = [  
    df["Billing_Amount"] > 30000,  
    (df["Billing_Amount"] > 10000) & (df["Billing_Amount"] < 30000),  
    df["Billing_Amount"] < 10000  
]  
  
choices = ["High Fees", "Medium Fees", "Low Fees"]  
  
df["Billing_Category"] = np.select(conditions, choices, default="Unknown")
```

[43]

```
df["Billing_Category"]
```

[44]

```
...  
0      Medium Fees  
1      High Fees  
2      Medium Fees  
3      High Fees  
4      Medium Fees  
...  
54961    Low Fees  
54962    High Fees  
54963    Medium Fees  
54964    High Fees  
54965    Low Fees  
Name: Billing_Category, Length: 54966, dtype: str
```

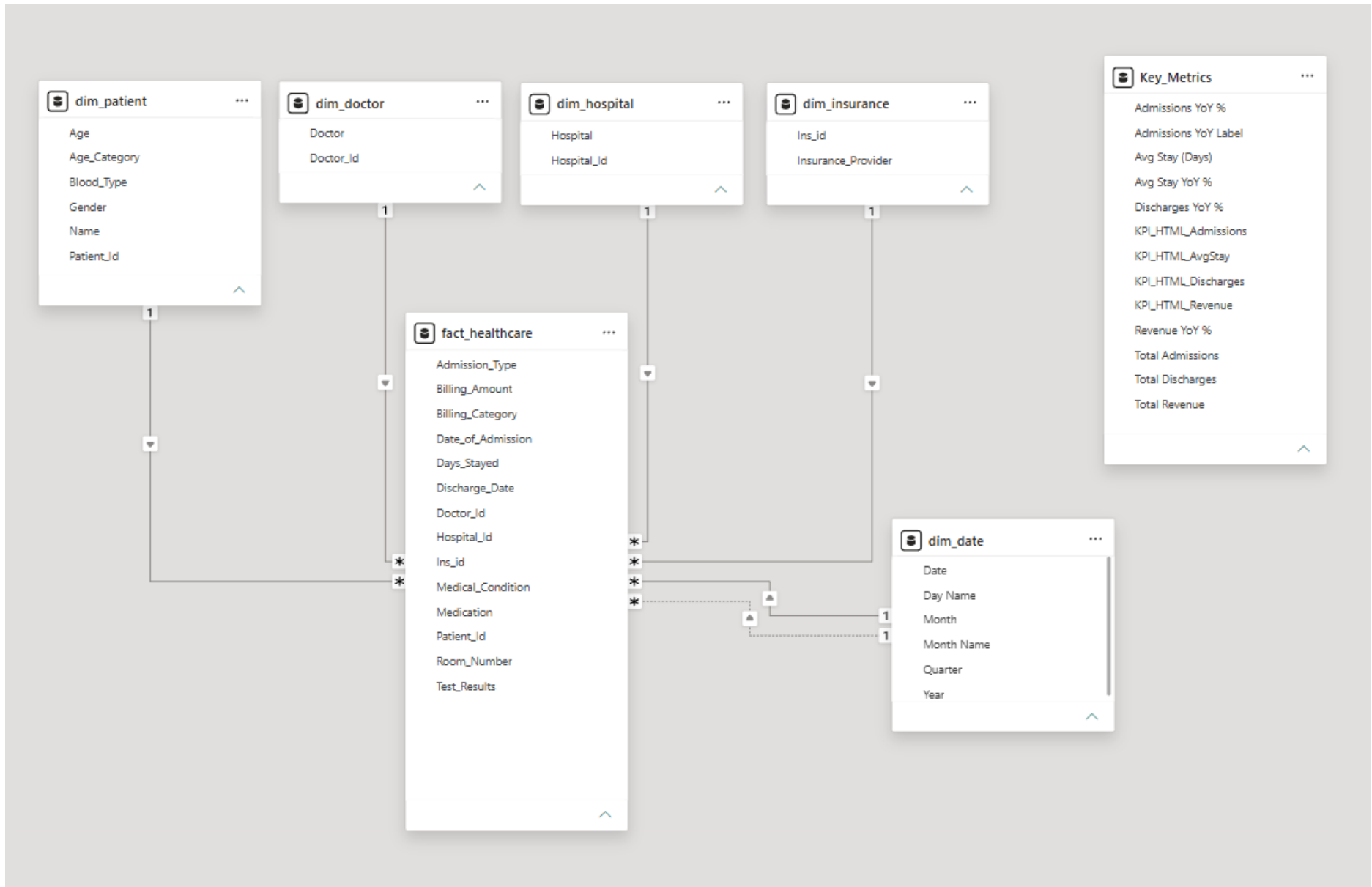
3. Creating a New "Age_Category" column based on Column "Age"

```
age_conditions = [  
    df["Age"] < 18,  
    (df["Age"] >= 18) & (df["Age"] <= 35),  
    (df["Age"] >= 36) & (df["Age"] <= 50),  
    (df["Age"] >= 51) & (df["Age"] <= 65),  
    df["Age"] > 65  
]  
  
age_choices = ['Pediatric',  
               'Young Adult',  
               'Middle-Aged Adult',  
               'Pre-Senior',  
               'Geriatric']  
  
df["Age_Category"] = np.select(age_conditions, age_choices, default="n/a")
```

```
df["Age_Category"]
```

```
0      Young Adult  
1      Pre-Senior  
2      Geriatric  
3      Young Adult  
4      Middle-Aged Adult  
...  
54961  Middle-Aged Adult  
54962  Pre-Senior  
54963  Middle-Aged Adult  
54964  Middle-Aged Adult  
54965  Pre-Senior  
Name: Age_Category, Length: 54966, dtype: str
```

4. Power BI Data Model



5. Power BI DAX

These are the key Dax I used for this project

1. Total Revenue

```
Total Revenue = SUM(Healthcare_Admissions[Billing_Amount])
```

2. Total Admissions

```
Total Admissions = COUNTROWS(Healthcare_Admissions)
```

3. Average Length of Stay (ALOS)

```
Avg Length of Stay = AVERAGE(Healthcare_Admissions[Length_of_Stay])
```

4. Revenue MoM Growth %

```
Revenue Growth % =  
VAR PriorRevenue = CALCULATE([Total Revenue], DATEADD('Calendar'[Date], -1, MONTH))  
RETURN  
DIVIDE([Total Revenue] - PriorRevenue, PriorRevenue, 0)
```

5. Payer Revenue Share %

```
\  
  
Payer Revenue Share % =  
DIVIDE(  
    [Total Revenue],  
    CALCULATE([Total Revenue], ALL(Healthcare_Admissions[Insurance_Provider])),  
    0  
)
```

6. Dashboard



Healthcare Operations & Financial Overview

Patient Throughput • Clinical Drivers • Hospital Performance

Clear all filters



Date

All



Hospital

All



Doctor

All



Insurance Provider

All



Age

All



Blood Type

All



TOTAL ADMISSIONS

54,966

↑ 20.3% vs. previous period



TOTAL REVENUE

\$1,404.1M

↑ 20.4% vs. previous period



AVG. LENGTH OF STAY

15.5 Days

↑ 0.1% vs. previous period

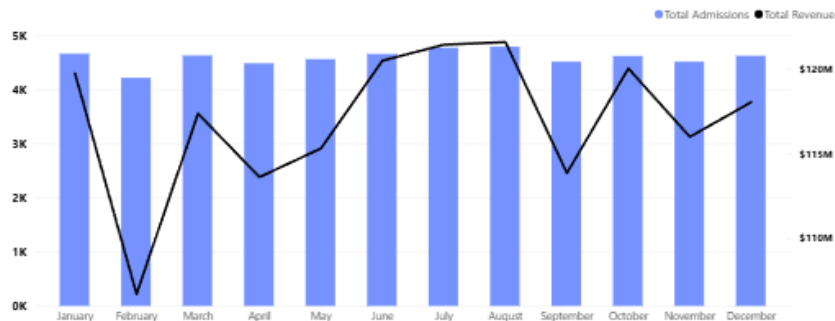


TOTAL DISCHARGES

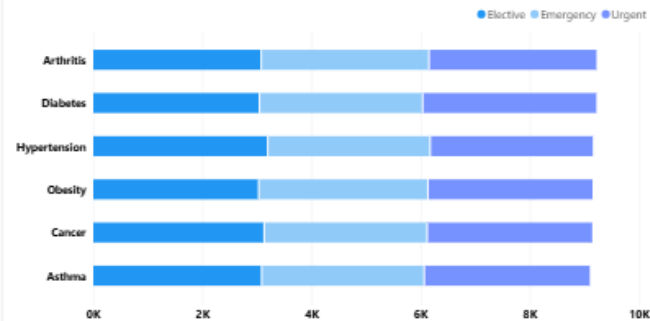
55.0K

↑ 21.6% vs. previous period

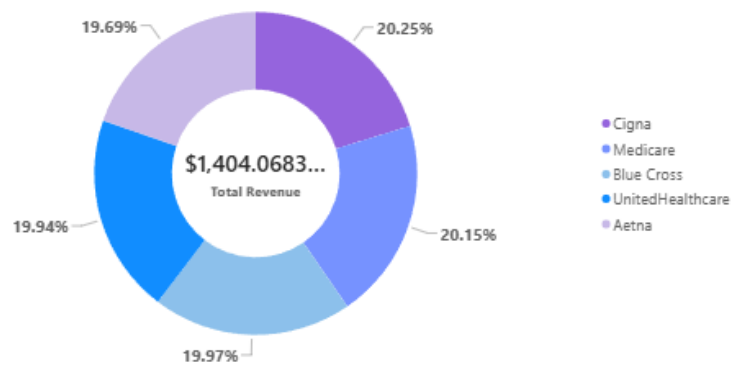
Monthly Admissions & Revenue Performance



Top Clinical Drivers (by Patient Volume)



Revenue Share by Insurance Provider



Patient Details

Name	Gender	Blood Type	Age	Admission Type	Sum of Billing Amount
Aaron Adams	Female	O-	38	Elective	26,052.11
Aaron Aguirre	Male	A-	36	Emergency	27,087.56
Aaron Anderson	Female	A+	50	Urgent	39,804.66
Aaron Anderson Md	Female	A-	20	Elective	16,846.42
Aaron Archer	Female	B-	47	Urgent	10,602.08
Aaron Archer	Female	B-	49	Urgent	10,602.08
Aaron Baker	Female	A+	84	Emergency	6,826.68
Aaron Baker	Male	B+	73	Elective	10,135.89
Aaron Baldwin Jr.	Male	O-	20	Urgent	29,740.96
Aaron Barnes	Male	O-	85	Urgent	4,783.44
Aaron Bautista	Male	B-	21	Urgent	49,827.52



Data driven decisions. Healthier communities

7. Strategic Recommendation

- Optimize Average Length of Stay (ALOS) & Discharge Pipelines

At 15.5 days, the current ALOS limits patient throughput and bed turnover. Implement early discharge planning, post-acute care coordination, and daily multidisciplinary rounds to safely reduce stay duration by 1–2 days. This can increase bed availability for elective admissions without requiring physical expansion.

- Align Bed & Resource Allocation with High-Volume Conditions

Diabetes, Cancer, and Hypertension represent some of the highest admission volumes across emergency and elective channels. Aligning bed allocation and nursing schedules with condition-specific demand patterns can reduce emergency department waiting times and improve operational efficiency.

- Streamline Claims & Accelerate Reimbursement Cycles

Revenue is distributed relatively evenly across the major insurance providers. Dedicated payer-specific claim audit teams can monitor reimbursement delays, improve pre-authorization processes, reduce claim denials, and accelerate cash flow.

- Implement Demand-Forecasted Staffing Models

Monthly patient volume and billing figures show seasonal fluctuations that affect staff workload. Using demand-based staffing models instead of static schedules can ensure adequate physician and nursing coverage during peak periods while helping control overtime costs.

- Standardize Clinical Protocols & Reduce Billing Variance

Billing amounts and length of stay vary among physicians treating similar conditions. Benchmarking workflows against top-performing doctors can help standardize care pathways, reduce unnecessary diagnostic testing, and control cost-per-case variation across facilities.